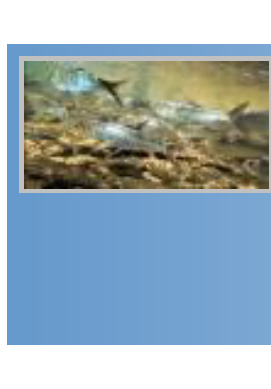


Alosa pseudoharengus
(Alewife)
Fishes
Native Transplant

Collection Info
Point Map
Species Profile
Animated Map

Image 1 of 3

U.S. Fish and Wildlife Service

Alosa pseudoharengus (Wilson, 1811)

Common name: Alewife

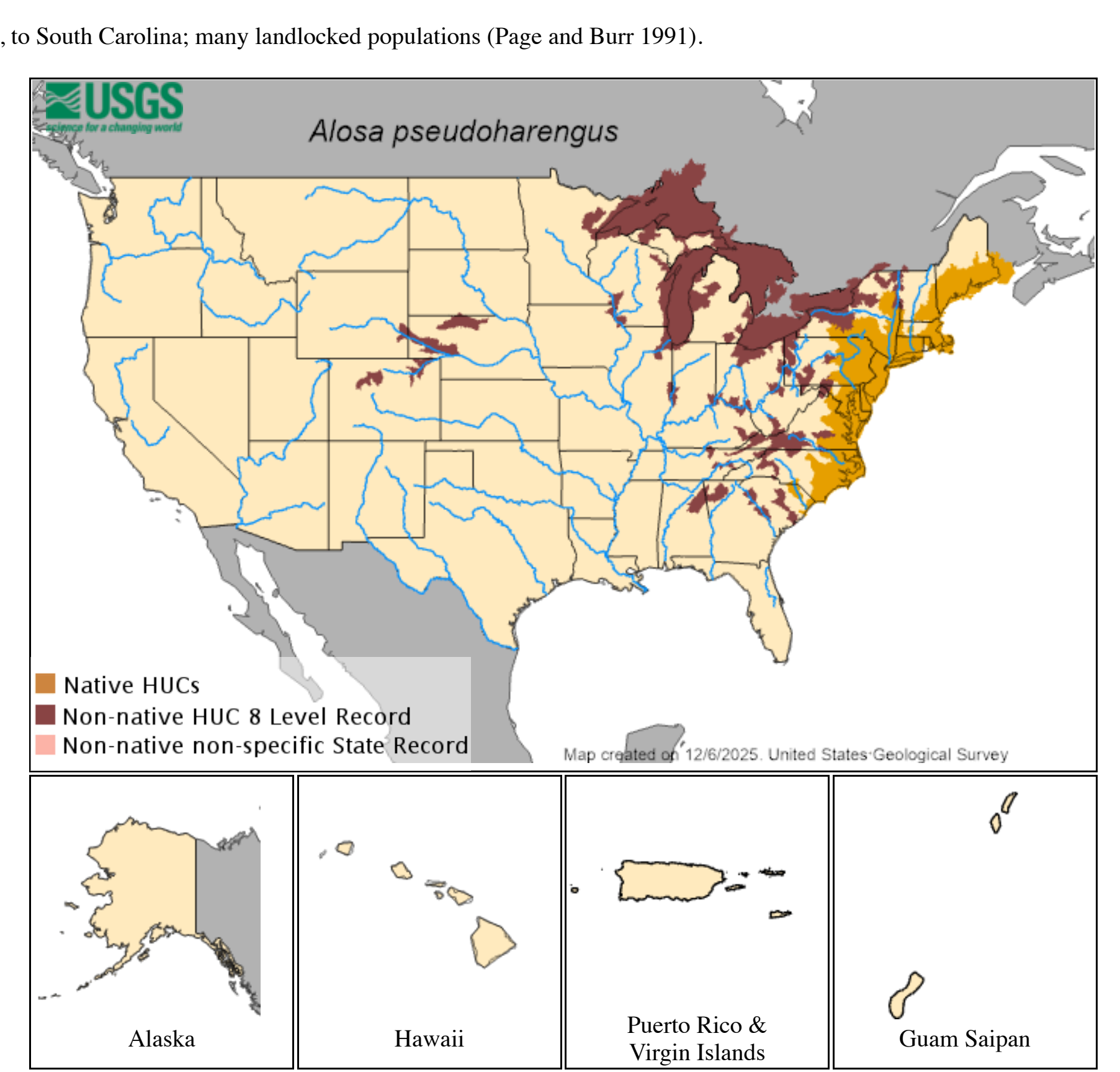
Synonyms and Other Names: mulhaden, grey herring, golden shad, kyak, sawbelly, grayback, river herring

Taxonomy: available through ITIS

Identification: The Alewife is a small herring with a dark dorsal side, bluish to greenish, and light sides with horizontal darker stripes. The head is broad and triangular and the body is relatively deep. Eyes are large with adipose eyelids. A dull black spot is located behind the operculum. Scales are easily rubbed off and form scutes on the midline of the belly. Jaw teeth are inconspicuous and tongue teeth are absent. Caudal fin is forked and lacks an adipose fin. Alewife are visually similar to Blueback Herring (*Alosa aestivalis*), but Alewife has a white peritoneal lining, larger eyes, and a greater body depth than Blueback Herring (Whitehead 1985; Page and Burr 1991; Etnier and Starnes 1993; Jenkins and Burkhead 1994; Scott and Crossman 1998).

Size: Total length up to 38 cm, but inland populations are usually less than 25 cm. Fertilized eggs have a diameter around 0.9 mm and larvae are on average 3.8 mm long at hatch (Henrich 1981).

Native Range: Atlantic Coast from Red Bay, Labrador, to South Carolina; many landlocked populations (Page and Burr 1991).


Alosa pseudoharengus
Native HUCs
Non-native HUC 8 Level Record
Non-native non-specific State Record
Map created on 12/6/2025. United States Geological Survey
Alaska
Hawaii
Puerto Rico & Virgin Islands
Guam Saipan

Native range data for this species provided in part by NatureServe
Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Alosa pseudoharengus* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AL	2018	2018	1	Middle Coosa
CO	1973	2006	2	Colorado Headwaters; Middle South Platte-Sterling
GA	1971	2019	4	Coosawattee; Etowah; Middle Savannah; Upper Coosa
IL	1949	2022	6	Chicago; Des Plaines; Lake Michigan; Little Calumet-Galien; Pike-Root; Upper Illinois
IN	1956	2014	3	Lake Michigan; Little Calumet-Galien; Middle Wabash-Busseron
KY	1986	2000	4	Little Scioto-Tygarts; Middle Ohio-Laughery; North Fork Kentucky; Ohio Brush-Whiteoak
ME	1985	2007	2	Lower Kennebec River; New England Region
MD	1979	1979	1	Youghiogheny
MI	1933	2025	19	Betsie-Platte; Betsy-Chocolay; Carp-Pine; Detroit; Fishdam-Sturgeon; Keweenaw Peninsula; Lake Erie; Lake Huron; Lake Michigan; Lake St. Clair; Lake Superior; Muskegon; Ontonagon; Pere Marquette-White; Raisin; St. Clair; St. Marys; Sturgeon; Waiska
MN	1956	2017	4	Baptism-Brule; Beaver-Lester; Lake Superior; St. Louis
NE	1965	1997	4	Lower North Platte; Middle Niobrara; Middle North Platte-Scotts Bluff; Snake
NH	1985	1985	1	New England
NY	1868	2020	19	Black; Chaumont-Perch; Chenango; Headwaters St. Lawrence River; Hudson-Hoosic; Irondequoit-Ninemile; Lake Champlain; Lake Erie; Lake Ontario; Lower Genesee; Niagara River; Oak Orchard-Twelvemile; Raisin River-St. Lawrence River; Salmon-Sandy; Saranac River; Seneca; St. Regis; Upper Hudson; Upper Susquehanna
NC	1999	1999	1	Upper Catawba
OH	1931	2024	9	Ashtabula-Chagrin; Black-Rocky; Cedar-Portage; Chautauqua-Conneaut; Huron-Vermilion; Lake Erie; Ohio Brush-Whiteoak; Sandusky; Shenango
PA	1931	2012	11	Bald Eagle; Chautauqua-Conneaut; Conemaugh; Connoquenessing; Conococheague-Opequon; French; Lake Erie; Lower Susquehanna-Penns; Shenango; Upper Ohio; Youghiogheny
SC	1971	2009	6	Congaree; Cooper; Lake Marion; Upper Catawba; Upper Savannah; Wateree
TN	1991	1993	4	Obey; Upper Clinch, Tennessee, Virginia; Watauga, North Carolina, Tennessee; Watts Bar Lake
VT	1997	2023	3	Lake Champlain; Mettawee River; Richelieu
VA	1965	2010	10	Appomattox; Banister; Kanawha; Middle New; Roanoke; Roanoke Rapids; Upper Dan; Upper Levisa; Upper New; Upper Roanoke
WV	1967	1998	4	Little Muskingum-Middle Island; Lower New; Middle New; Upper Kanawha
WI	1944	2018	13	Beartrap-Nemadji; Coon-Yellow; Door-Kewaunee; Grant-Little Maquoketa; La Crosse-Pine; Lake Michigan; Lake Superior; Lower Fox; Manitowoc-Sheboygan; Milwaukee; Peshtigo; Pike-Root; St. Louis

Table last updated 12/8/2025

† Populations may not be currently present.

Ecology: Alewife is anadromous and euryhaline (prefers <15 psu), but occurs in landlocked water bodies including the Great Lakes (DiMaggio et al. 2016). It exists at various depths ranging throughout the year from littoral to profundal zones depending on the season. In spring, Alewife spreads out across a lake, staying in warmer waters above the thermocline in schools and dispersing shoreward at night to spawn near the surface of open lake shores, bays, harbors, and lower reaches of rivers (O'Gorman et al. 2013). When near shore, Alewife prefers rocky substrates to sandy substrates (Janssen and Kuebke 2004). During summer, young-of-year (YOY) Alewife stays in the warmer epilimnion while older fish will often venture into the thermocline and cooler waters (Wells 1968; Otto et al. 1976; Johannsson and O'Gorman 1991). In fall, Alewife moves away from shore and into deeper waters as the thermocline descends and weakens and eventually overwinters in the profundal zone. Near the end of winter, Alewife begins to move shoreward once again to repeat the cycle (Wells 1968; Bergstedt and O'Gorman 1989).

Reproduction of Alewife is polygynandrous and lasts for around a month in spring once water temperatures surpass ~15ºC (Edsall 1970; Hlavek and Norden 1987). Landlocked populations generally mature a year faster than anadromous populations, taking two years for males and three for females. Females deposit between 10,000 and 360,000 eggs (typically on the lower end) at random on any type of substrate (Hlavek and Norden 1987; Scott and Scott 1988; Scott and Crossman 1998). Eggs are non-adhesive and incubation time varies with temperature, from 15 days at 7.2ºC to 3.7 days at 21.1ºC (Edsall 1970). Larvae are phototropic, pelagic, and begin feeding two days after hatching (Odell 1934; Heinrich 1981).

Alewife can survive in temperatures between 3 to 31ºC, but prefer waters between 16 to 20ºC (Otto et al. 1976; Spotila et al. 1979; Dufour et al. 2008). Water above 31ºC and below 3ºC can cause extreme stress and eventual death (Colby 1973; Otto et al. 1976; McCauley and Binkowski 1982). Lepak and Kraft (2008) found that Alewife experienced sublethal immunosuppression when held in ponds below 2ºC for six weeks. Die-offs of Alewife often occur in the Great Lakes and are typically concurrent with severe winter conditions and/or when fish conditions are poor due to high population densities and increased disease transfer rates (Brown 1972; Colby 1973; Bergstedt and O'Gorman 1989; Lepak and Kraft 2008). Winter temperatures sustained at or below 1ºC can cause osmoregulatory failure due to changes in lipid composition leading to fish mortality. Hence, smaller Alewife with small lipid reserves often experience higher winter mortality than their larger counterparts (Snyder and Hennessy 2003). Therefore, YOY Alewife must be at least 60 mm in total length to successfully overwinter in the Great Lakes, especially in Lake Superior where the growing season is shorter and winter is harsher (Brown 1972; Elrod 1983; O'Gorman and Schneider 1986). As a consequence to the climate in Lake Superior, Alewives there are much fewer in number and much longer than others of the same age in the other Great Lakes due to extreme selection pressure for fast growth (O'Gorman et al. 1997). The warming effects of climate change in Lake Superior are expected to increase the favorable environment for Alewife by reducing the lethality of winter and by promoting food production (Bronte et al. 2003; Hook et al. 2007; O'Gorman et al. 2013).

The diet of Alewife is diverse and impactful, as it is a proficient feeder of eggs, insects, zooplankton, and larval fish. It selects for the largest zooplankton in invaded water bodies and subsequently has altered the species composition of zooplankton in the Great Lakes where Alewife is abundant (Hutchinson 1971; Johannsson et al. 1991). The degree to which Alewife reduces large zooplankton populations is so severe that the seasonal movements of the fish in Lake Ontario have been tracked by the size of zooplankton (O'Gorman et al. 1991). Alewife also preys on larval fish, benefited by its wide-ranging seasonal movements and ability to feed in midwater and low light levels (O'Gorman et al. 2013). Alewife's predation of fish whose larvae are pelagic can be so severe that their recruitment is drastically reduced (Madenijan et al. 2008). A variety of piscivorous fish consume Alewife in the Great Lakes. Alewife is vital prey for both nearshore (e.g. Yellow Perch (*Perca flavescens*) and Walleye (*Sander vitreus*)) and offshore (e.g. Rainbow Smelt (*Osmerus mordax*), Burbot (*Lota lota*), and Salmonid) predators (O'Gorman 1974; Elrod et al. 1981; Ridgway et al. 1990).

Means of Introduction: There is apparently disagreement concerning the native status of Alewife in Lake Ontario. Miller (1957) and Smith (1970) point out the first record from Lake Ontario was in 1873. Smith (1970) is of the opinion that it was introduced into the lake. Although Smith (1970) brings up the possibility that Alewife were introduced into Lake Ontario with American Shad stockings in the 1880s, he discounts this possibility in favor of the hypothesis that they reached the lake via the Erie Canal from the Hudson River. He contends that Alewife was only able to invade the lake after the decline of predators such as Lake Trout and Atlantic Salmon in the 1860s. Other authors believe this species was probably native to Lake Ontario (Lee et al. 1980 et seq.) and spread through the Great Lakes via the Welland Canal (Lee et al. 1980 et seq.). The species was first reported from Lake Erie in 1931, Lake Huron in 1933, Lake Michigan in 1949, and Lake Superior in 1954. The spread of Alewife in the upper Great Lakes is thought to be enabled by warmer weather in the 1950's (O'Gorman and Stewart 1999). The Alewife was intentionally stocked in inland waters. The population in the New River, West Virginia, resulted from stockings in Claytor Lake, New River, Virginia (Jenkins and Burkhead 1994). The recently discovered population in Lake St. Catherine, Vermont, is likely a result of an illegal stocking (Good, personal communication). Lakes in the Adirondack Mountains and Otsego Lake, New York were illegally stocked with Alewife for forage (Smith 1985; Sinnott, personal communication; D. Warner, personal communication).

Status: Introduced populations have been established in 22 US states, the Canadian province of Ontario, and throughout the Great Lakes. Introduction to the Youghiogheny River was unsuccessful (Hendricks et al. 1979).

Great Lakes:
Widespread, with populations reproducing and overwintering at self-sustaining levels in all five Great Lakes. However, current populations in the lower Great Lakes have severely declined from peak abundances throughout the 1900s due to salmonid stockings, dreissenid mussel invasion, and food web shifts (O'Gorman et al. 2013).

Impact of Introduction: Presence of the Alewife could restructure a lake's food web, leaving less food for native species (USEPA 2008). Disappearance of native planktivorous salmonids, such as whitefish, in the Great Lakes has been attributed in part to the introduction of Alewife, which reduced zooplankton populations (Crowder and Binkowski 1983; Todd 1986; Page and Laird 1993). Crowder (1984) speculated that a cisco native to Lake Michigan, the bloater *Coregonus hoyi*, evolved fewer and shorter gill rakers, and shifted to benthic habitat and diet as a result of competition with Alewifes. Smith (1970) attributed the extermination of the lake herring and decline of chub species in the Great Lakes to the Alewife. Smith also talks about the various interrelated changes that took place in each of the Great Lakes as Alewife abundance increased. Christie (1972), on the other hand, argues that the Alewife was not responsible for these changes. The Alewife is the dominant fish in Lake Michigan. It accounts for 70–90% of the fish weight (Becker 1983). Alewife has recently become the dominant prey item for double-crested cormorants in Lake Champlain (DeBruyne et al. 2012). Additionally, Alewife (both age-0 and adults) show significant spatial overlap with age-0 rainbow smelt (*Osmerus mordax*) in Lake Champlain, which could alter population dynamics of both species through competition (between age-0 Alewife and smelt) or predation (by adult Alewife on age-0 of both species) and limit the availability of these forage fish to larger predators such as lake trout (*Salvelinus namaycush*) and Atlantic salmon (*Salmo salar*) (Simonin et al. 2012). Pothoven et al. (2013) documented an increased abundance of age-0 yellow perch (*Perca flavescens*) and changes in the zooplankton community structure in Saginaw Bay, Lake Huron following the disappearance of Alewife, including increased abundance of cladocerans (e.g., *Daphnia* spp., *Bythotrephes*, *Leptodora*) and decreased abundance of cyclopoid copepods. Alewife contain high levels of thiaminase, which reduces absorption and assimilation of thiamin in predators such as salmonids can cause reduced body condition, swim performance, and other potenial impacts (Houde et al. 2015)

Alewife is a very important species in the history of biological invasions in the Great Lakes. Periodic large-scale die-offs littered the beaches of the Great Lakes with rotting fish in the 1960's. Such die-offs can pose both a nuisance and a health hazard (Becker 1983). Prompted by calls for Alewife management, Pacific salmonids were introduced to both control Alewife populations and utilize Alewife as a food source for sport fisheries.

Remarks: Although there is a report of two small Alewife taken from the Colorado River, Texas (Bean 1882), we believe this record is in error. Bean (1882) reported that the specimens were sent to Professor Baird at the National Museum. However, a query of the museum's holdings did not return these specimens. We believe the fish are more likely either misidentified *A. chrysochloris* or *A. sapidissima*. *Alosa sapidissima* were stocked in the Colorado River in 1874 (Bean 1882).

Alewife is one of the most frequently found prey items in the diet of the Double-Crested Cormorant in the southern basin of Lake Michigan (Madura and Jones 2016).

Voucher specimens: Michigan (UMMZ 157215, 160969, 167872, 171308, 170945), Wisconsin (UMMZ 162861, 167945).

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Other Resources:
Alternative Strategies for the management of non-indigenous alewives in Lake St. Catherine, Vermont (pdf)

Map of locations in Illinois ILNHS

Alosa spp. (ANS Clearinghouse Bibliography)

GLIFWC-Maps

Great Lakes Water Life

Global Invasive Species Database Factsheet

US Fish and Wildlife Service Ecological Risk Screening Summary for *Alosa pseudoharengus*

FishBase Summary

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